

Thesis evaluation

submitted at the Faculty of Mathematics and
Physics of Charles University

☒ supervisor's report

☐ reviewer's report

☐ bachelor's thesis

☒ master's thesis

Author: Michal Valentík

Thesis title: Resonant particle production of an evolving oscillating scalar field

Study program and field of study: Theoretical physics

Year of submission: 2026

Supervisor's name and titles: Dražen Glavan, dr.

Affiliation: CEICO, Institute of Physics (FZU) of the Czech Academy of Sciences

Contact email: glavan@fzu.cz

Academic standard of thesis:

☒ excellent ☐ very good ☐ average ☐ below average ☐ unsatisfactory

Factual errors:

☒ hardly any ☐ corresponding to thesis extent ☐ numerous minor ☐ major

Results:

☒ original ☐ original and non-original ☐ non-trivial compilation ☐ cited from references
☐ copied

Extent of thesis:

☒ large ☐ standard ☐ sufficient ☐ insufficient

Graphical, language and formal standards:

☒ excellent ☐ very good ☐ average ☐ below average ☐ unsatisfactory

Typos:

☒ hardly any ☐ number corresponding to thesis extent and topic ☐ numerous

General thesis standards:

☒ excellent ☐ very good ☐ average ☐ below average ☐ unsatisfactory

Supervisor's assessment, comments and suggestions:

As a supervisor of the Student's master's thesis project, I can say that the Student has performed admirably in all aspects of the research project, from the initial familiarization with and learning of the methods to the later stages of applying the developed methods to examples and writing the thesis.

The emphasis in the project was on technical details and on developing mathematical approximations appropriate for describing more realistic parametric resonance regimes in the cosmological era of preheating. This included extending the current analytic approximations to account for the non-harmonic evolution of the oscillating inflaton. Non-harmonic evolution is a necessity in the inflaton's evolution, because the inflaton oscillations have to decay into the matter that our Universe is made of. Thus far, semi-analytic approximations including such effects have not been worked out. The derivation of such approximations in this thesis therefore presents an original contribution to the literature, and the plan is to summarize the thesis results and write them up as a research article in the near future.

The Student has demonstrated a higher-than-average level of independence in this technically rather challenging project. The thesis itself is very well and clearly written. It first gives an overview of the current state-of-the-art semi-analytic approximation methods that describe resonant particle production in the narrow-resonance regime. It then proceeds to develop and generalize these methods to describe situations where the inflaton's amplitude and frequency evolve slowly in time, leading to an effective smearing of the standard resonance-band picture. The methods and approximations are derived systematically and pedagogically, by considering simpler cases first and generalizing to more complicated cases later. The developed approximations are compared to exact numerical results, demonstrating regimes where the approximations work very well and identifying the limitations of the approximations where they become less reliable. The resulting approximations can best be summarized by a first-order set of equations for the amplitude of the produced field perturbations, which are much easier to solve on a computer, as they capture only effects on long time scales, while fast oscillations are solved for analytically.

For these reasons, I emphatically suggest the grade "excellent" for the Student's master's thesis research project and the submitted thesis.

Optional questions for the defense and topics for discussion:

The thesis is quite thorough in covering the cosmological motivation and in developing the methods for describing parametric resonance with slow deviations from harmonic driving. I would ask whether the Student can envision any direct practical applications of the methods and equations derived in this thesis.

I

☒ recommend

☐ do not recommend

to accept the submitted work as a master's / bachelor's thesis.

I suggest the thesis to be graded as:

☒ excellent ☐ very good ☐ satisfactory ☐ failed

Place, date and signature of supervisor:

Dražan Glavan
Prague, 01.06.2026.

